

How Our View of Space Changed S6E1a

A **model** is a simpler way to show something hard to see. When scientists find **new evidence**, models change — that's a *strength* of science, not a weakness!

GEOCENTRIC (old idea)

"Geo" = **Earth**. Earth sits still at the **center**; the Sun, Moon & stars circle around it. It *looks* true from the ground — but it's wrong.

HELIOCENTRIC (modern view)

"Helio" = **Sun**. The **Sun is at the center** and the planets orbit it. Proposed by **Copernicus**; proven when **Galileo's telescope** saw moons orbiting Jupiter — not everything circles Earth!

★ **Big Idea:** Better tools (the telescope) → new evidence → the model changed. Science updates when new information appears.

Our Place in the Universe S6E1b

Earth → Solar System → Milky Way Galaxy → Universe

Level	What It Is	Helpful Comparison
Earth	The planet we live on	One house
Solar System	The Sun + everything orbiting it (8 planets, moons, asteroids, comets)	Your neighborhood
Milky Way Galaxy	A huge group of billions of stars , including our Sun	Your entire city
Universe	Everything — all galaxies, stars, and space that exists	The whole planet Earth

Our Sun is just **one star** of hundreds of billions in the Milky Way — and the Milky Way is just **one galaxy** of billions in the universe.

ACTIVITY 1: Zoom Out!

Number these from **1 (smallest)** to **5 (largest)**:

____ Milky Way Galaxy
____ Earth

____ Universe
____ Solar System

____ The Moon

Meet the Planets S6E1c

8 planets orbit the Sun: the 4 closest are **small & rocky**; the 4 farthest are **giant balls of gas and ice**.

Planet	Type	What Makes It Special
Mercury	Rocky	Closest to the Sun; smallest planet; almost no atmosphere
Venus	Rocky	Hottest planet — thick poisonous atmosphere traps heat
Earth	Rocky	The only planet known to support life ; liquid water on the surface
Mars	Rocky	The "Red Planet"; largest volcano in the solar system
Jupiter	Gas Giant	Largest planet; giant storm = the Great Red Spot
Saturn	Gas Giant	Famous bright, wide rings made of ice and rock
Uranus	Ice Giant	Spins on its side ; pale blue-green color
Neptune	Ice Giant	Farthest ; very cold, with the fastest winds

Mercury

Venus

Earth

Mars

— belt —

Jupiter

Saturn

Uranus

Neptune

🧠 **Mnemonic:** "My Very Excellent Mother Just Served Us Nachos" = order from the Sun. ⚠️ **Pluto?** In 2006 scientists made clearer rules — Pluto is now a **dwarf planet**. (Another model that changed!)

ACTIVITY 2: Name That Planet!

- Hottest planet: _____
- Has the Great Red Spot: _____
- Spins on its side: _____
- Only planet with known life: _____
- The "Red Planet": _____
- Famous for bright rings: _____

🦋 Flip over for the habitable zone, gravity & inertia, and comets, asteroids & meteoroids! →



Comparing Planets & the "Just Right" Zone S6E1c

To compare planets, look at **size vs. Earth**, **distance from the Sun**, and **ability to support life**.

Too Close = Too Hot 🔥

Mercury & Venus: scorching heat, no liquid water. Venus is hot enough to melt lead!

Habitable Zone ✅

Earth sits "just right" — liquid water, breathable air, the right temperature. Every living thing we know needs **liquid water**.

Too Far = Too Cold ❄️

Gas/ice giants: freezing, **no solid surface** to stand on. Mars: too cold, air too thin (but scientists study it closely).



Why Planets Orbit: Gravity & Inertia S6E1d

GRAVITY — the pull ←

A force that **pulls objects toward each other**. More **mass** = stronger pull. The Sun is gigantic, so its gravity pulls planets toward it.

INERTIA — keep going straight →

The tendency of a moving object to **keep moving in a straight line**. Inertia "wants" to send each planet flying straight off into space.

★ **Together:** Gravity pulls **inward** + inertia pushes **straight ahead** = a curved path called an **ORBIT**. **Picture this:** a ball on a string — your hand pulling in = gravity; the ball wanting to fly off = inertia. Let go (no gravity) and it flies straight off!



ACTIVITY 3: Gravity or Inertia?

Write **G** (gravity) or **I** (inertia) for each:

- | | |
|---|---|
| _____ 1. Pulls planets toward the Sun | _____ 4. Would send Earth flying straight off if the Sun vanished |
| _____ 2. Keeps a moving object going in a straight line | _____ 5. The "hand pulling the string" in the ball demo |
| _____ 3. Gets stronger when an object has more mass | _____ 6. The reason the ball flies off when you let go |



Comets, Asteroids & Meteoroids S6E1e

Object	Made Of	Where It's Found	Key Feature
Comet	Ice, dust & rock ("dirty snowball")	Far edges of the solar system	Grows a glowing tail as it nears the Sun
Asteroid	Rock and metal	Mostly the asteroid belt (between Mars & Jupiter)	Rocky leftovers from the early solar system
Meteoroid	Small piece of rock or metal	Anywhere; often broken off asteroids or comets	Becomes a "shooting star" if it burns up in our atmosphere

⚠️ **One rock, three names!** In space = **meteorOID** → glowing streak in the sky = **meteOR** → lands on the ground = **meteorITE**.
Memory trick: "-ite" hit the ground!



ACTIVITY 4: Name That Space Object!

- | | |
|---|---|
| 1. Icy object that grows a tail near the Sun: _____ | 5. Space rock that lands on Earth's surface: _____ |
| 2. Rock found between Mars and Jupiter: _____ | 6. Ring-shaped region between Mars & Jupiter: _____ |
| 3. Small rock floating in space: _____ | |
| 4. The glowing streak ("shooting star"): _____ | |



I Can... (Self-Check)

- | | |
|--|--|
| ☐ Explain how the geocentric and heliocentric models differ | ☐ Explain why Earth can support life (habitable zone) |
| ☐ Explain why scientific models change (new evidence!) | ☐ Describe how gravity + inertia create orbits |
| ☐ Order Earth → Solar System → Milky Way → Universe | ☐ Tell apart comets, asteroids & meteoroids (and meteor/meteorite!) |
| ☐ Name the 8 planets in order from the Sun | |
| ☐ Compare rocky planets and gas/ice giants | |